Digital Applications in Archaeology and Cultural Heritage 41 (2026) e00537

Contents lists available at ScienceDirect

Digital Applications in Archaeology and Cultural Heritage

journal homepage: www.elsevier.com/locate/daach

Digital heritage preservation through drone surveying: Vulnerability analysis of Venetian church façades

Irene Rocca [a,*], Giuseppe D'Acunto [a], Ilaria Forti [a], Anna Saetta [a,b]

[a] DCP, University IUAV of Venice, Terese Campus, Dorsoduro 2206, 30123, Venezia, Italy
[b] Associate to National Research Council (CNR), Institute for Construction Technologies (ITC), Italy

ARTICLE INFO

Keywords:
Digital heritage preservation
Drone-based surveying
3D digital modeling
Façade vulnerability
Digital Twin
Cultural Heritage conservation

ABSTRACT

Preservation of Venice's unique cultural identity requires innovative approaches to safeguard its fragile architectural heritage. This study explores the potential of drone-based surveying and Historical Digital Twin methodologies to assess the vulnerability of city's façade systems. Focusing on emblematic case studies, high-resolution drone imagery was employed to generate detailed 3D digital models of the façades and their statuary elements. Models served as base for structural vulnerability analyses, enabling precise monitoring, maintenance planning, and safety evaluations. The research demonstrates how the integration of advanced digital surveying techniques with structural engineering expertise can enhance understanding and management of risks affecting Venice's cultural heritage. This research integrates advanced information technologies into the broader context of design, engineering, and maintenance of artistic buildings and assets, demonstrating how digital tools can revolutionize cultural heritage conservation. The findings offer a replicable framework that bridges traditional preservation methods with cutting-edge digital modeling techniques to enhance the management of risks.

1. Introduction

Since the second half of the last century, the need and urgency for the affirmation and recovery of the places' own identity through the care, preservation and enhancement of the Cultural Heritage, has been recognized nationally and internationally. Venice represents an exemplary case due to the exceptional concentration of historic architecture, the close relationship between buildings and the lagoon environment, and the specific vulnerabilities affecting its built heritage. The peculiarities and intrinsic critical issues of this city require specific choices to be made for its preservation. The subject of this work regards the monumental façades, which represent a significant case of how much their protection and enhancement affects the perpetuation of the city's identity.

The Venetian façade system thus represents one of the main identity features of the lagoon city and, at the same time, it constitutes an extremely fragile ensemble, especially because of its structural and material characters in relation to the lagoon environment (Settis, 2016; Di Sipio and Zezza, 2011). The present research work constitutes a first stage of a more ambitious project that promises to map the Venetian façade system through its most representative examples. A mapping of the main Venetian façade systems that, by means of a digital survey both overall and in detail, will constitute a real atlas capable of documenting both their historical-artistic values and their construction peculiarities, highlighting their fragility and conditions of degradation.

In particular, the paper will focus on two case studies: the façades of church Santa Maria di Nazareth and church San Francesco della Vigna, considered by the authors to be emblematic for several reasons and, above all, for the construction characteristics that make them particularly representative of two cases of strong vulnerability in the panorama of the Venetian façade system. For both case studies, the digital survey was carried out with a well-established methodology: the use of photogrammetry, combining drone with terrestrial tools, and laser scanning (Gonizzi Barsanti et al., 2022; Gonizzi Barsanti and Guidi, 2018). In particular, digital images acquirement by means of drones for the survey of buildings or sites of cultural heritage and related digital model development by "Structure-from-Motion" (SfM) techniques gained more and more importance in the last years and an extensive body of literature has developed on the subject (Azzola et al., 2019).

Drones (or unmanned aerial vehicles - UAVs) have proven to be invaluable in the fields of archaeology and cultural heritage, as they provide a noninvasive, time- and cost-efficient way to document and

* Corresponding author.
E-mail addresses: irocca@iuav.it (I. Rocca), giuseppe.dacunto@iuav.it (G. D'Acunto), ilaria.forti@iuav.it (I. Forti), saetta@iuav.it (A. Saetta).

https://doi.org/10.1016/j.daach.2026.e00537
Received 6 August 2025; Received in revised form 7 March 2026; Accepted 20 April 2026
Available online 21 April 2026
2212-0548/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

monitor cultural heritage sites (Themistocleous, 2020). In addition to high-resolution data acquisition over large areas in a short period of time, drone photogrammetry enables rapid generation of 3D digital models for documentation and reconstruction. These models, thanks to the technological development that in recent years has allowed a considerable advance in the digitization systems of architectural assets and also the creation of platforms for the visualization of unequivocal 3D data, make it possible to create a database and a tool at the service of the entities responsible for cultural heritage protection and also to make the cultural heritage more accessible.

As a starting point for defining appropriate working methodologies, Federman's studies (Federman et al., 2017) describe a procedure for acquiring data useful to SfM techniques elaboration. For the direct geo referencing of pictures by drone's sensors, Gabrlik's work can be mentioned (Gabrlik, 2015). A comparison is made between traditional aerial photogrammetry and photogrammetry by drones by Bolognesi, as well as how they can be integrated (Bolognesi et al., 2014). Among the research works that focus the investigation specifically on drone survey of cultural heritage assets, Barrile's paper describes the adopted methodology for elaborating a 3D model of the Greek Walls of Reggio Calabria starting from the images acquired by drones (Barrile et al., 2017), Azzola uses UAV photogrammetry and consequent modeling for the preservation and mapping of venetian walls of Bergamo (Azzola et al., 2019), Themistocleous uses its potentials for Asinou church in Cyprus (Themistocleous et al., 2015) and more recently Banfi for Appia Antica Archaeological Park (Banfi et al., 2023).

In both the selected case studies, given the construction specifics of the two façades, it was necessary to use the drone for filming the upper register of the façades. In fact, a high-precision survey, both overall and in detail, of the upper tier was a prerequisite for the success of the research work, for problems otherwise of strong distortions and approximations. The statues and decorative apparatus placed in the higher parts of the structures are very articulated and, at the same time, represent the most fragile and vulnerable elements because they are most exposed to kinetic stresses of various kinds, and it is precisely for this reason that the evaluation of their vulnerability is the subject of the research work. This process leading to the vulnerability assessments of cultural heritage assets (here San Francesco della Vigna and Santa Maria di Nazareth façades of Venice city) is a topic still little treated in the literature.

2. Adopted methodology: from historical digital twin to vulnerability assessment

This study employs a survey methodology based on drone flights to create historical digital twins of constructions within Cultural Heritage, with a specific focus on the city of Venice, an iconic site in the Italian landscape. These digital models are designed to provide valuable tools for Cultural Heritage management by administrative authorities, as well as to support the development of structural models for assessing the vulnerability of these assets to various potential hazards. These include seismic events and the effects of high tides (“acque alte”) that frequently impact Venice—a challenge further intensified by climate change.

The first phase of the procedure involves planning the drone survey operations, including defining flight trajectories and assessing restrictions imposed by no-fly zones in certain airspaces. Following the drone survey, the results are integrated with a supplementary phase of ground-based surveys and processing, utilizing more traditional methods (laser scanning and static photogrammetry).

Upon completion of the survey, the integrated workflow moves to data processing, and then to the development of a three-dimensional solid digital model, hereinafter referred to as Historical Digital Twin (HDT) of the asset of Cultural Heritage under study (Marra et al., 2021). This particular application builds upon the Digital Twin concept, which was first conceptualized by (Glaessgen and Stargel, 2012) and was subsequently applied to built structures by [(Angjeliu et al., 2020; Xiaozhu and Xingwang, 2018; Jouan and Hallot, 2019), amongst others] as a digital counterpart of their physical reality. This stage involves a multi-faceted approach, encompassing SfM techniques, automated reconstruction processes (including alignment, dense point cloud and meshing generation), and manual reconstruction, complemented by the utilization of reverse engineering software. The procedure, theorized by the authors in (Rocca et al., 2024), is here outlined in Section 4.

Having a digital model (HDT) provides multiple opportunities, from asset management and enhancement through the implementation of detailed historical, critical, and material analyses, to the development of structural models supporting vulnerability assessments. Unlike standard 3D models, the HDT serves as an interoperable platform that merges geometric precision with historical stratigraphy and material decay data. Indeed, the issue of deterioration can be addressed through mapping and analyses, which are subsequently incorporated into the graphic restitution of the asset. Moreover, the methodology hopes to bridge the gap between geomatics and structural engineering through a specialized reverse engineering phase. By transitioning from discrete mesh data to continuous NURBS surfaces, the study then enables the generation of accurate solid Finite Element models (FEM), minimizing geometric uncertainties that typically affect the structural simulation of complex Cultural Heritage shapes.

Of particular significance, the proposed procedure is applicable to any scale of Cultural Heritage constructions, ranging from entire buildings to specific components (e.g., a monumental façade) or even individual objects (such as individual high-relief statues or movable artistic assets). A specific novelty of this research is indeed the focus on the seismic vulnerability of projecting decorative elements, a segment of cultural heritage that is typically under-researched from a structural safety perspective. The methodology can address themes as behaviour under self-weight loads, out-of-plane phenomena and the amplification of seismic action at different heights of structures, directly linking high-resolution surveying with kinematic and structural analyses.

3. Venice case study: UAV photogrammetric survey of church monumental façades and statues

As an example of applying the proposed methodology for the construction of historical digital twins at two scales—building element and artistic element—this work focuses on the monumental façades of two Venetian churches and some statues that characterize them, positioned at various heights: the church of “Santa Maria di Nazareth” (vulgo degli Scalzi), Fig. 1 and the church of “San Francesco della Vigna”, Fig. 2.

Due to the nature of the objects under study, constructing models that both accurately represent a digital twin of the elements and serve as the basis for structural models used in vulnerability assessments necessitates a survey campaign relying predominantly on drones.

This approach is necessary to accurately survey the tops of the façades, as the limited depth of field would otherwise result in significant distortions. For example, in the case of Santa Maria di Nazareth façade, the distance from the Grand Canal is only about 10 m. Relying solely on terrestrial photogrammetry would produce a highly inclined framework, leading to inaccuracies in the model, shaded areas, and uncontrolled interpolations between data points.

Similarly, for the San Francesco della Vigna façade, the issue of limited depth of field arises due to the presence of a building located just 5 m in front of most of the façade. As with the first example, the distortions and shadowed areas caused by using terrestrial instruments only would be substantial and incompatible with the level of reliability required for the 3D model.

3.1. Historical background

The monumental façades of the analyzed churches are two of the most emblematic representations of Venetian identity. The first, Santa Maria di Nazareth (commonly known as the Scalzi Church), serves as a



(a) (b)

Fig. 1. Church of Santa Maria di Nazareth (or degli Scalzi): (**a**) the façade (ph. G. De Vecchi), (**b**) the location.



(a) (b)

Fig. 2. Church of San Francesco della Vigna: (**a**) the façade, (**b**) the location.

prominent "landmark," recognizable since the late 19th century as a gateway to the city of Venice. The second, San Francesco della Vigna, is probably Andrea Palladio's first ecclesiastical project in Venice and the only one fully executed under his direction.

The Church of Santa Maria di Nazareth, designed by Baldassarre Longhena and completed with a Baroque façade by Giuseppe Sardi (1680), represents a unique iconographic landmark of the late Venetian Baroque. Located near Venice's Santa Lucia station, it is exposed to significant anthropic pressure and a harsh lagoon environment characterized by saline mist, tidal fluctuations, and extreme thermal cycles. Uniquely for Venice, the façade is constructed entirely from Carrara marble (Bettini and Frank, 2014), a rare lithotype in the Venetian context. Its complex geometry - rich in niches, recesses, and statuary - creates microclimates that promote differential interaction of the marble support with rainwater and salts. Over time, this saccharoid limestone has proven vulnerable, as evidenced by a 2013 incident involving the detachment of a capital's acanthus leaf. Subsequent diagnostics revealed severe degradation, including decohesion and pulverization reaching depths of up to 12 cm. Consequently, a conservative restoration project of this façade (Rocca et al., 2022) was initiated, prioritizing experimental, compatible, and reversible interventions. Beyond structural recovery, the project established a comprehensive technical-scientific database to monitor the efficacy of these actions and provide a framework for future maintenance. Particular focus was required by the façade decorative-statuary apparatus. The Carrara marble statuary (Fig. 3a) indeed remains a critical point due to its fragility, as evidenced by the historical loss of the 'Speranza' statue in 1920 (SABAPV-AS).

Designed by Andrea Palladio on commission from Patriarch Grimani and completed in 1564, the façade of San Francesco della Vigna is a seminal example of Venetian Renaissance architecture and an experimental application of Palladian compositional principles. Clad entirely in Istrian stone, the façade's composition reflects the interior spatial hierarchy, with a central nave and lower side chapels, through a tripartite system of a central tympanum and lateral semi-tympanums.



(a) (b)

Fig. 3. Statue of (**a**) San Girolamo of Santa Maria di Nazareth, (**b**) Mosè of San Francesco della Vigna.

Historically, the structure has faced structural problems rather than material deterioration. Differential foundation settlement at the left corner has induced a slow outward rotation, necessitating continuous monitoring and restoration of the capitals, the most vulnerable architectural elements.

Providing a chromatic contrast to the pale Istrian stone are the monumental bronze statues of Moses and St. Paul by Tiziano Aspetti (Fig. 3b) in the niches of the façade. These elements suffered advanced corrosion: thick sulfate encrustations and white staining have developed along rainwater run-off planes, reaching critical levels of degradation.

3.2. Survey program

Regarding Santa Maria di Nazareth, a few years after the completion of the restoration work, it was decided to use this monumental façade as the first case study for developing the methodology described in Section 2. This methodology, based on extensive data acquisition, aimed to produce an historical digital twin representing the monument, facilitating a comprehensive understanding of its vulnerabilities and enabling ongoing safety assessments. This methodology was also applied to the façade of San Francesco della Vigna.

To address vulnerabilities in both the structural integrity and material conservation of these façades and their artifacts, a comprehensive survey campaign was planned with the support of the VIDE laboratory (part of IR.IDE research infrastructure) at IUAV University of Venice, leading to the creation of the façade's digital twin through 3D modeling.

A 3D digital photogrammetry survey served as the basis for creating solid models used to assess static and seismic performance. As previously mentioned, the most suitable method for conducting an accurate survey under conditions like those of the two façades, required an intensive and strategic use of drone systems, complemented by some ground-based survey operations. For the aerial photogrammetric survey, a drone DJI Mavic 2 Pro with High Dynamic Range (HDR) video 4k 10 bit was employed, with characteristics explicated in Table 1. The choices made for the photos taken by the drone are: exposure ISO 100, aperture f/5.6, exposure time 1/400s.

Table 1
Characteristics of Mavic 2 Pro drone.

Sensor size	25.4x25.4 mm
Image size	5472 × 3648 pixels
Focal length	28 mm
Flight time	31 min

In order to check the correctness and accuracy of the photogrammetric elaborations some relevant morphological points were selected on the façade (e.g., building corners) from a network of trilaterate key points, along with strategically distributed Ground Control Points (GCPs) consisting of clearly recognizable artificial targets. GCPs were homogeneously distributed across the façade to ensure comprehensive coverage, positioning them at the corners, at the center, and at various heights. Their coordinates were acquired using a total station to establish an absolute topographic reference. Subsequently, the GCPs were manually identified across multiple oriented images within the processing software to optimize the model's internal and external orientation.

The façade of Santa Maria di Nazareth measures about 22.4m in width, 24.0m in height, while San Francesco della Vigna façade measures 32m in width and 28.0m in height.

Drawing upon experience from previous artifacts, to ensure a sufficient point density and data to represent the actual state of the artifacts by means of a 3D model at a 1:20 drawing scale, the model must be capable of reaching 1 cm accuracy. As a consequence the ground sampling distance (GSD) has to be equal to or below 1 cm/pixel. The theoretical value of the mean distance of acquisition to get the desired resolution is referred to an ideal condition, with a camera placed on a tripod without external interferences (Banfi and Mandelli, 2021); considering instead the real conditions of execution of the survey in both cases the authors decided for a flight at a mean distance of 2 m from the asset. Consequently, an effective GSD of 0.7 mm/pixel was obtained. This accounts for potential motion blur and environmental instabilities (e.g., wind gusts and aerodynamic interference near the façade) that may affect pixel sharpness during flight. Consequently, the number of images collected was remarkable, almost 700 photos at the maximum resolution of the camera for every façade. They are gathered in JPG format and are comprehensive of the coverage also of decorative elements and of statues in niches and on top.

As regards the planning of the flight path, this was carried out in manual flight manner, programmed by the authors and carried out by

the pilot, following regular paths as much as possible. For the survey of the façade, horizontal strips were carried out, from bottom to top, see Fig. 4a–b for the specific path on the two façades. Instead, for surveys of the statues exposed on the sides or on the top, 360-degree helical flights were performed to capture the complexity of the shapes from different views and not to have shadow areas, see Fig. 4c.

In particular, the choice of flight altitude aimed at dividing the façade of S. Maria di Nazareth into 5 strips and San Francesco della Vigna one into 4 stripes, plus any additional photos for particular elements of the decorative-statuary apparatus, such as the statues. The vertical overlap was evaluated directly on the screen by the pilot of the drone, while the transversal overlap (interval acquisition) was set such as to have two-thirds of the previous photo in the next one. Thus, the photos (Fig. 5) were arranged in a sequence that was certainly onerous and at times redundant, but such that it met the requirements of obtaining a sufficiently dense point cloud in the 3D model construction software.

The defined acquisition parameters allow the same flight path to be replicated in future monitoring campaigns, enabling consistent comparisons and the assessment of degradation over time.

3.3. Support and role of terrestrial survey

The 3D survey of the façade systems was carried out also through other indirect techniques: laser scanning and static photogrammetry, both from a terrestrial point of view. In particular, the instruments employed are the FARO focus S 150 terrestrial laser scanner and the Nikon D850 digital camera with 50 mm f/1.4 lens, whose specifications are given in Table 2.

This survey was not conducted to merge the data from the photogrammetric flight with the terrestrial photogrammetric data. Instead, the two techniques played distinct, complementary roles in the overall process. The Terrestrial Laser Scanner (TLS) was used to support the UAV survey by providing a high-accuracy contextual survey and a corresponding model of the surrounding environment, including the church and the area around the façade. This served as a stable topographic backbone to support the UAV survey. Regarding the close-range terrestrial photogrammetry, it served as a metric benchmark for the lower portions of the façades. By capturing these portions at a higher level of detail, it enabled a comparative validation of the UAV-derived model, ensuring consistency between the aerial and ground-based datasets. To guarantee spatial coherence across all platforms, the same GCPs of the aerial photogrammetric survey have been taken in the terrestrial survey, ensuring a unified coordinate system for the final comparative analysis.

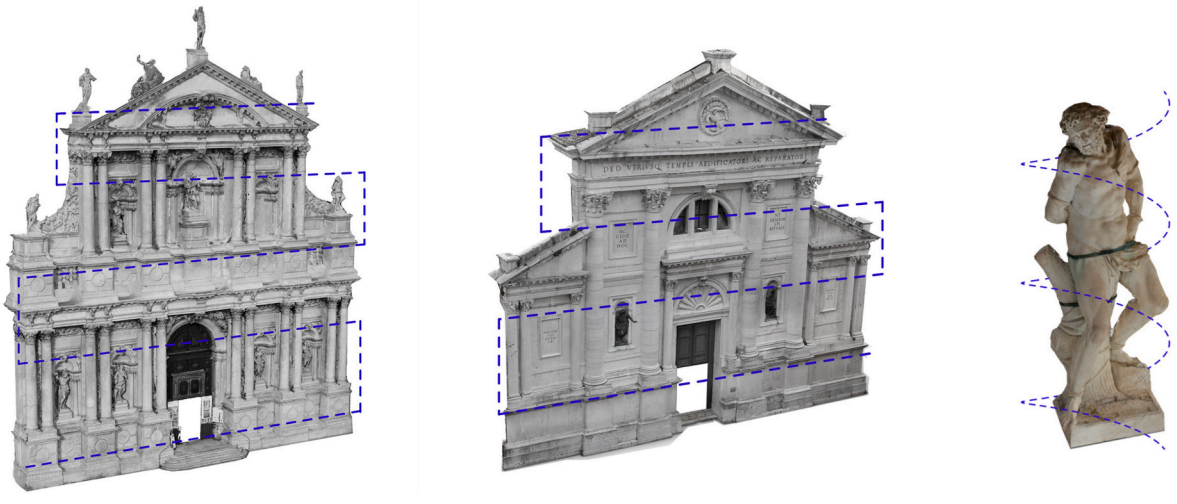
For the photographic survey, the authors followed a shooting path similar to that of the drone, employing horizontal strips with comparable distances and overlaps. Nevertheless, data acquisition was restricted to the lower half of the façade to mitigate excessive image distortion in the higher sections. Regarding the laser scans, five scans were conducted at distances of 10 to 20 m from each façade. For Santa Maria di Nazareth, the scans were taken from the quay in front of the church and from the Scalzi Bridge, the highest accessible point near the church. For San Francesco della Vigna, the scans were taken from the square in front of the church and from the terrace of a building located to the left of the façade.

4. 3D modeling and digital twin elaboration

The models were developed primarily using Agisoft Metashape Pro (version 2021, (Metashape AgiSoft PhotoScan)). All UAV survey images were imported into the software. The automatic photo modeling process employs the SfM detection technique, which uses computer vision algorithms to identify significant points in individual photographs, derives their parameters, cross-reference identifiable points across multiple images, and determines their spatial coordinates through nearly automatic collimation (Lucignano, 2021). During the image orientation phase, a sparse point cloud is generated. The subsequent phase involves extracting a dense point cloud, geo-referencing, scaling the model, and manually refining it to exclude noise or points unrelated to the analyzed geometry.

Next, a 3D photorealistic textured polygonal mesh is built over the dense cloud, creating three-dimensional objects based on parameters aligned with a three-dimensional coordinate system, effectively capturing both the texture, shape and geometry of reality.

The 3D modeling process then continues with the reconstruction of the façade system and its individual defining elements, completing the 3D elaboration derived from the survey described in Section 3. The solid model, which accurately represents the correct volumes, is generated from the mesh model exported from Agisoft Metashape Pro in OBJ format (along with its high-definition texture). This data is subsequently processed using the reverse engineering software Rhinoceros (McNeel Rhinoceros). Unlike a purely visual 3D shell, a volume-representative mesh ensures topological integrity, allowing for the precise calculation



(a) (b) (c)

Fig. 4. Regular manual flight path followed for (**a**) façade of Santa Maria di Nazareth, (**b**) façade of San Francesco della Vigna, (**c**) San Bartolomeo statue, belonging to Santa Maria di Nazareth façade.



Fig. 5. Drone in action and close-up images from drone survey.

Table 2
Characteristics of terrestrial survey instruments.

FARO focus S 150		Nikon D850	
Reading speed	976000 points per s	Sensor size	35.9 x 23.9 mm
Accuracy (max)	1 mm at 150m	Image size	8256 x 5504 pixels
Accuracy (used)	2.1/10m	Focal length	50 mm
Field of vision	360° × 300°		

of structural metrics such as displacement, mass distribution, and volumetric decay. The resulting three-dimensional solid digital model is referred to as Historical Digital Twin (Marra et al., 2021), framework introduced in Section 2. The precision and detail of the survey directly enhance the accuracy and quality of the 3D model rendering.

The full procedure for survey acquisition and 3D restitution for cultural heritage preservation is thoroughly detailed in (Rocca et al., 2024).

4.1. Façades

The acquired survey data were processed and optimized to produce reality-based three-dimensional digital models of the façades (see Fig. 6). These figures demonstrate that by prioritizing precision and survey quality, a model with an average error of less than 1 cm for control points and distances was achieved. This level of accuracy is compatible with the objectives of the 3D model, which include control, measurement, detection of material and structural vulnerabilities, color alterations, cracks, damages, and the creation of a reliable model for subsequent numerical analyses.

The workflow continued with mesh model processing to generate the solid 3D model, which also accurately represents the correct volumes. Due to complexity, diversity of shapes, and scale of these façades, this process could not be fully automated and required manual reconstruction of the volumes in Rhinoceros. Classical architectural elements (such as columns and lintels) were modeled by extracting sections and elevations directly from the 3D model or the point cloud.

As an example, Fig. 7 illustrates the workflow for the digital reconstruction of the Historical Digital Twin of the Santa Maria di Nazareth façade, transitioning from the photorealistic 3D model to the solid one.

4.2. Statues

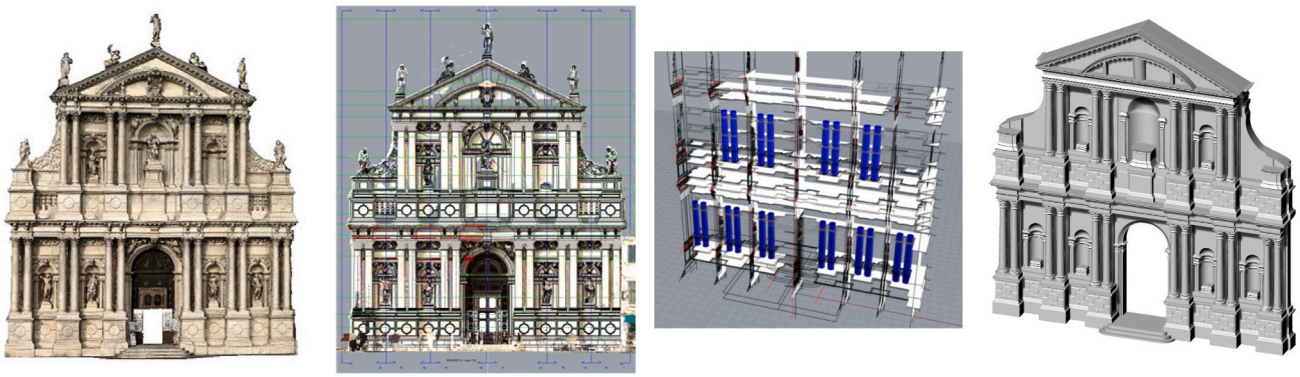
The complex geometries, such as those of the statuary apparatus (and other decorative elements like the shield), were processed separately from the entire façade to create high-resolution models of these elements independently of the overall architectural structure. This process can be conducted in two ways: either by isolating the point cloud around the decorative element within the global point cloud obtained from photo alignment or by using pertinent photos captured during the UAV survey focused on the statue.

The 3D modeling of the statues followed the same procedure used for the entire façade, as illustrated in Fig. 8 with the example of the statue of San Girolamo. Unlike classical architectural elements, statues and other decorative components are freeform structures, requiring a different approach for solid 3D model reconstruction. In these cases, the transition from the textured mesh model to the solid one is semi-automatic and includes generating NURBS models as well as manual corrections by the operator. These corrections address common mesh errors such as non-manifold edges, intersections, redundant faces, and other irregularities. In particular, in the statues' case, in the final step of converting the surface mesh into a solid model, the methodology employs a reverse engineering software (McNeel Rhinoceros) for transitioning from a mesh



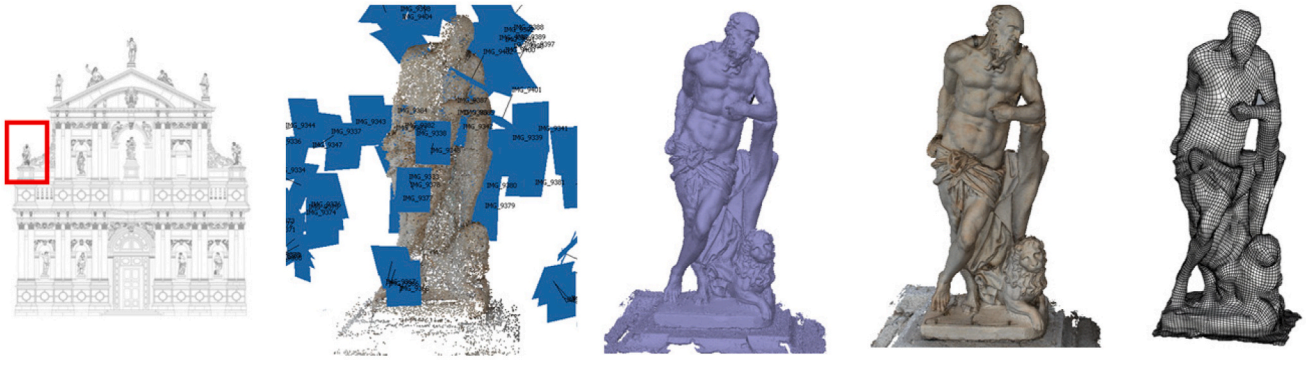
(a) (b)

Fig. 6. 3D digital model of the façade of (**a**) Santa Maria di Nazareth, (**b**) San Francesco della Vigna.



(a) (b) (c) (d)

Fig. 7. Workflow of Scalzi Façade modeling: (**a**) 3D digital model of the surfaces, (**b**) and (**c**) reconstruction phases in Rhinoceros, (**d**) solid model representative also of the correct volumes.



(a) (b) (c) (d) (e)

Fig. 8. San Girolamo statue: (**a**) location, steps of 3D modeling - (**b**) sparse cloud with shooting points, (**c**) dense cloud, (**d**) textured mesh model – (**e**) mesh model representative also of the correct volumes.

to a NURBS surface, which is then exported to the 3D geometric modeler (MIDAS FEA NX) to generate a solid finite element (FEM) model. This 3D

digital model, which accurately represents the correct volumes, can be effectively used for both in-depth static and seismic FEM analyses and for expeditious assessments.

5. Towards safety assessment

The obtained 3D model is used to identify vulnerabilities from a kinematic perspective, including the presence of crack patterns, detachments, chromatic alterations, deterioration of reinforcements and restraints introduced during restoration interventions (e.g., steel supports for statues or protective nets). It also allows the detection of potential deficiencies in the performance of these interventions. For instance, the drone survey enables the evaluation of static risk factors, such as cracks, material detachments, and movement evolution (i.e., kinematics) in the upper parts of the façade and its projecting elements, including the coat of arms and statues. This information supports effective planning of potential direct interventions. Additionally, such factors may also serve as indicators of preservation risks in the façade's texture and integrity.

Thanks to the high precision of the 3D model, technicians with specialized engineering expertise can quickly identify critical points and interpret the structural behavior of the façade. Among the issues verified using the 3D model is the inclination of the upper portion of the façade of San Francesco della Vigna. An out-of-plumb condition was identified through verticality verification by analyzing thin slices of points (from the dense mesh or the model itself), reconstructing the tympanum's geometry and comparing it with the vertical axis.

Using the three-dimensional façade model, it is possible to understand the role of individual components and use it as the basis for creating a numerical-mathematical model (detailed finite element 3D modeling, FEM) of the façade or its elements for static and dynamic structural safety analyses. The development of FEM models from SfM survey is a topic of considerable interest in literature, see, for instance, (Castellazzi et al., 2017; Gonizzi Barsanti and Guidi, 2017).

The model enables assessments of static and seismic vulnerabilities of the structure as a whole and of its constituent parts at different levels of detail. These analyses account for mechanical actions, such as wind, impacts, seismic forces, as well as for the effects of environmental variations typical of the lagoon area. Specifically, the high-precision 3D digital model supports safety checks using either simplified (Berto et al., 2021) or advanced methods (e.g., FEM, in this study performed with MIDAS Gen software (MIDAS Gen)). Depending on the required level of analysis detail, the historical digital twin can provide parameters for simplified methods or serve as a basis for computational FE models for in-depth static and dynamic evaluations.

5.1. Entire façade of Santa Maria di Nazareth

For example, focusing on the entire façade of Santa Maria di Nazareth, a numerical FE model was generated as the final step of the reconstruction and analysis process. Fixed constraints were applied at the base of the façade, accounting for the deep pile foundations. The mechanical properties of marble and masonry were assumed considering the evidence from the literature (Italian Standards (MIT, 2019)) and the findings of the diagnostic campaign conducted during the most recent restoration, resulting in: (i) for the Carrara marble elastic modulus E equal to 60000 MPa, compressive strength ranging between 31 and 72 MPa and tensile strength 2–4.5 MPa respectively, having considered a material factor $\gamma_m = 2$ and, on the safe side, a confidence factor FC = 1.3; (ii) for the masonry behind elastic modulus E equal to 1050 MPa - corresponding to a cracked masonry, Poisson ratio 0.2, weight density 18 kN/m^3, compressive strength ranging between 0.6 and 1 MPa, assuming a material factor $\gamma_m = 3$ and a confidence factor FC = 1.3.

A preliminary self-weight load analysis was conducted using MIDAS Gen, with the results in terms of displacements and effective stresses shown in Fig. 9. This analysis highlights the primary structural issues affecting the decorative apparatus before accounting for dynamic actions. Another safety check involves analyzing the stress distribution across the entire façade to ensure that tensile and compressive stresses remain within the marble's material strength limits. For further details on these analyses, refer to the authors' dedicated work (Rocca et al., 2024).

5.2. Statuary apparatus of Santa Maria di Nazareth

Taking the statuary apparatus as an example, the generated 3D models and the collected data enable static and seismic vulnerability assessments of these elements. Despite being constructed of different materials—Carrara marble for the Santa Maria di Nazareth façade and bronze for San Francesco della Vigna—the statues share similarities in terms of analysis methodology. The different internal configurations (solid for marble and hollow for bronze) and material pathologies can be analyzed using the same approach. With the 3D model and its associated data, advanced methods (e.g., FEM) can evaluate stress concentrations and strain distributions, while simplified methods assess overturning tendencies under mechanical actions such as wind, shocks, or seismic forces. Fixed constraints were applied at the base of the statues, in order to reflect the actual physical connection between the statue and its pedestal, as evidenced by the Superintendency's archives and the data gathered during the survey program.

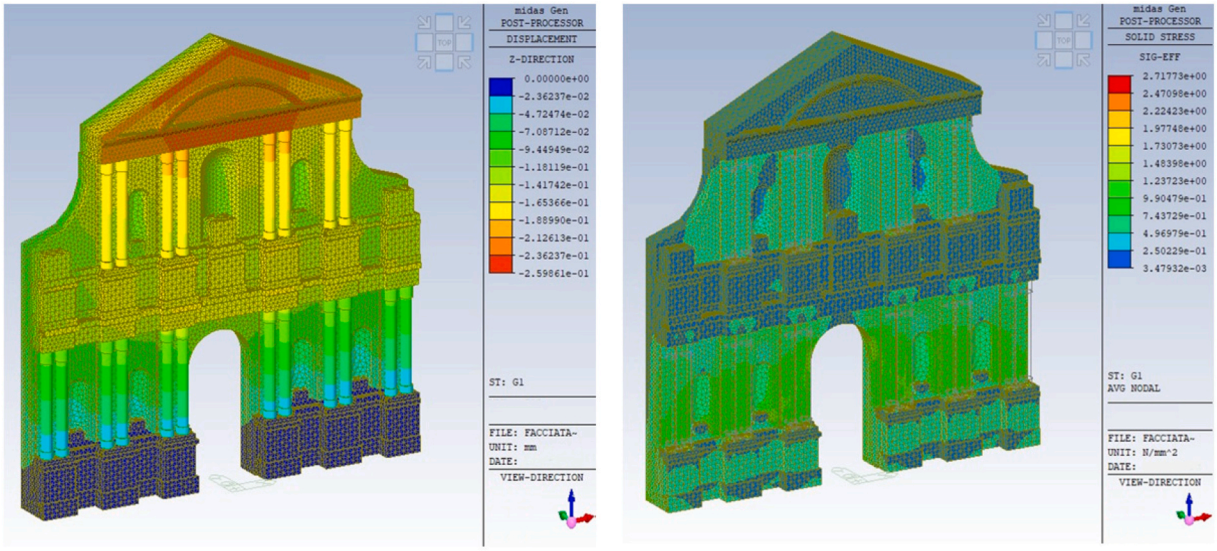
For example, Fig. 10 presents the results of the static analysis, preliminary to the seismic vulnerability analyses that require the evaluation of the seismic action at the base of the artifacts (see Rocca et al., 2024; MIT, 2019; Baggio et al., 2018), of the San Bartolomeo statue's numerical finite element model in terms of effective stresses. The analyses identify areas of maximum stress distribution, or zones potentially susceptible to damage in static conditions. Furthermore, they are a prerequisite for time-history simulations of potential strong motion events. The highest effective stresses on the San Bartolomeo statue are observed in the area connecting to the support trunk, which is attributed to the statue's pronounced eccentric center of gravity. However, the statue's safety under this load is confirmed, as the maximum stresses remain well within the Carrara marble's strength capacity.

6. Conclusions

Together with the classical methods of data collection—such as historical documentation and original 2D drawings to understand the construction techniques - this work demonstrates the utility of integrating advanced 3D surveys based on UAV digital photogrammetry to create a Historical Digital Twin of the monument and its apparatus. Through this process, the vulnerability assessments of the two case studies, the façades of San Francesco della Vigna and Santa Maria di Nazareth in Venice, have been carried out effectively and comprehensively.

The integration of survey data with advanced technologies and 3D representation, coupled with structural engineering expertise, represents a highly effective, precise, and accurate approach to gaining knowledge, preserving and ensuring the fruition of cultural heritage. In this study, this methodology was specifically tailored to the preservation and enhancement of the Venetian "façade systems", contributing to aims of addressing all aspects of artistic buildings and assets life cycle - from analysis and maintenance to advanced digital assessments and conservation strategies.

The acquired data were processed and optimized to generate a reality-based three-dimensional digital model of each façade, accurately capturing their geometric and morphological features. This model serves as a basis for identifying critical issues. Subsequently, a solid model, representative of both accurate geometry and correct volumes—referred to as the Historical Digital Twin—was developed from the initial mesh model. This Historical Digital Twin facilitates static and dynamic



(**a**) (**b**)

Fig. 9. Santa Maria di Nazareth façade FE model - results of linear analysis for self-weight loads (**a**) vertical displacements (in mm), (**b**) effective stresses (in MPa).

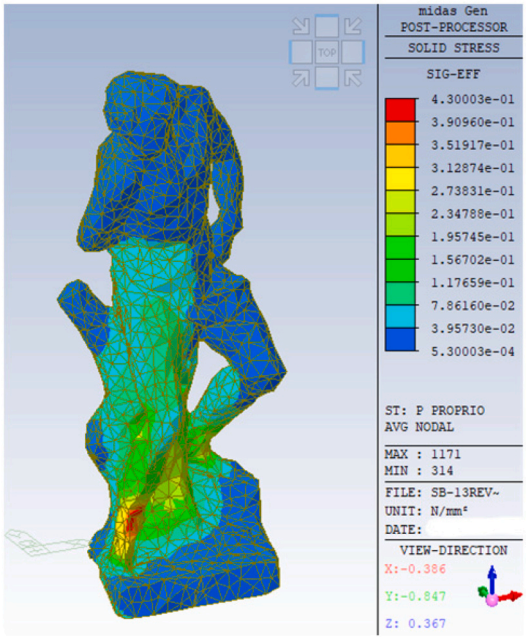


Fig. 10. San Bartolomeo statue FE model - effective stresses (in MPa) of linear analysis for self-weight loads.

analyses aimed at assessing the structural safety of the façade or its constituent elements.

The model's role as a "place of verification" underscores the importance of its high fidelity to real-world conditions, as this directly influences the reliability of subsequent evaluations. For monumental assets of this scale and complexity, such as the examined façade systems, achieving this level of accuracy would not have been possible through terrestrial surveys alone. Drone-based surveys were essential, especially for capturing the critical upper portions of the façades, where terrestrial methods would have resulted in distortions, shadowing, and gaps due to limited depth of field.

In summary, the critical and interpretative model obtained through this process not only serves as a database capable of describing the current state of conservation and identifying diverse issues but also ensures ongoing and effective monitoring when updated over time or integrated with new data. In particular, for the Scalzi façade, this tool can assist in developing a reasoned maintenance and monitoring plan, while for the façade of San Francesco della Vigna, it enables an assessment of its current condition to prioritize necessary conservation and restoration interventions.

The Historical Digital Twin is proposed as a tool for "preventive restoration" (concept elaborated in Cesare Brandi's Theory of Restoration (Brandi, 1963)), allowing for non-invasive monitoring and the simulation of future degradation scenarios to support administrative decision-making. Indeed, it enables analyses for safety assessments and analyses related to other risks, such as environmental conditions specific to the lagoon or the impact of weather over time. By situating this research within the broader scope of using information technologies to address challenges in construction technologies and management, this study underscores the transformative role of digital methods in bridging traditional conservation approaches with innovative engineering tools. This dual functionality—providing both documentation and analytical capabilities—makes it an invaluable tool for the preservation and safeguarding of cultural heritage.

CRediT authorship contribution statement

Irene Rocca: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Conceptualization. **Giuseppe D'Acunto:** Validation, Supervision, Project administration, Conceptualization. **Ilaria Forti:** Visualization, Resources, Project administration. **Anna Saetta:** Writing – review & editing, Validation, Supervision.

Data availability statement

Dataset available on reasonable request from the authors.

Funding

This research received no external funding.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Acknowledgments

The authors would like to thank Fr. Roberto Magni of O.D.C., Fr. Giampaolo Cavalli O.F.M. and Fr. Guido Ravaglia of O.F.M. for supporting the research, and Eng. Luca Grinzato and Francesca Castaldello Zamengo for their respective contribution.

References

Angjeliu, G., Coronelli, D., Cardani, G., 2020. Development of the simulation model for Digital twin applications in historical masonry buildings: the integration between numerical and experimental reality. Comput. Struct. 238, 106282. https://doi.org/10.1016/J.COMPSTRUC.2020.106282.

Azzola, P., Cardaci, A., Mirabella Roberti, G., Nannei, V.M., 2019. UAV photogrammetry for cultural heritage preservation modeling and mapping venetian walls of Bergamo. Int. Arch. Photogram. Rem. Sens. Spatial Inf. Sci. XLII-2/W9.

Baggio, S., Berto, L., Rocca, I., Saetta, A., 2018. Vulnerability assessment and seismic mitigation intervention for artistic assets: from theory to practice. Eng. Struct. 167, 272–286. https://doi.org/10.1016/j.engstruct.2018.03.093.

Banfi, F., Mandelli, A., 2021. Computer vision meets image processing and UAS PhotoGrammetric data integration: from HBIM to the eXtended reality project of Arco della pace in Milan and its decorative complexity. J. Imaging 7, 118. https://doi.org/10.3390/jimaging7070118.

Banfi, F., Roascio, S., Mandelli, A., Stanga, C., 2023. Narrating ancient roman heritage through drawings and digital architectural representation: from historical archives, UAV and LIDAR to virtual-visual storytelling and HBIM projects. Drones 7, 51. https://doi.org/10.3390/drones701005.

Barrile, V., Gelsomino, V., Bilotta, G., 2017. UAV and computer vision in 3D modeling of cultural heritage in Southern Italy. In: IOP Conference Series: Materials Science and Engineering, 225, 012196.

Berto, L., Meroi, E., Rocca, I., Saetta, A., 2021. Rocking activation of free standing elements in real conditions: a safe experimentally-based acceleration limit. Eng. Struct. 226, 111331. https://doi.org/10.1016/j.engstruct.2020.111331.

Bettini, G., Frank, M., 2014. La Chiesa di Santa Maria di Nazareth e la Spiritualità dei Carmelitani Scalzi a Venezia. Marcianus press, Venezia, pp. 24–25.

Bolognesi, M., Furini, A., Russo, V., Pellegrinelli, A., Russo, P., 2014. Accuracy of cultural heritage 3D models by RPAS and terrestrial photogrammetry. Int. Arch. Photogram. Rem. Sens. Spatial Inf. Sci. XL-5, 113–119.

Brandi, C., 1963. Teoria Del Restauro. Einaudi.

Castellazzi, G., D'Altri, A.M., de Miranda, S., Ubertini, F., 2017. An innovative numerical modeling strategy for the structural analysis of historical monumental buildings. Eng. Struct. 132, 229–248.

Di Sipio, E., Zezza, F., 2011. Present and future challenges of urban systems affected by seawater and its intrusion: the case of venice, Italy. Hydrogeol. J. 19, 1387–1401.

Federman, A., Santana Quintero, M., Kretz, S., Gregg, J., Lengies, M., Ouimet, C., Laliberte, J., 2017. UAV photogrammetric workflows: a best practice guideline. Int. Arch. Photogram. Rem. Sens. Spatial Inf. Sci. XLII-2/W5, 237–244.

Gabrlik, P., 2015. The use of direct georeferencing in aerial photogrammetry with Micro UAV. IFAC-PapersOnLine 48, 380–385.

Glaessgen, E., Stargel, D., 2012. The digital twin paradigm for future NASA and US air force vehicles. In: Paper Presented at the 53rd AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics and Materials Conference 20th AIAA/ASME/AHS Adaptive Structures Conference. 14th AIAA.

Gonizzi Barsanti, S., Guagliano, M., Rossi, A., 2022. 3D reality-based survey and retopology for structural analysis of cultural heritage. Sensors 22 (24), 9593. https://doi.org/10.3390/s22249593.

Gonizzi Barsanti, S., Guidi, G., 2017. A geometric processing workflow for transforming reality-based 3d models in volumetric meshes suitable for fea. Int. Arch. Photogram. Rem. Sens. Spatial Inf. Sci. XLII-2/W3, 331–338. https://doi.org/10.5194/isprs-archives-XLII-2-W3-331-2017.

GonizziBarsanti, S., Guidi, G., 2018. A new methodology for the structural analysis of 3D digitized cultural heritage through FEA. In: IOP Conference Series. Materials Science and Engineering 364-1, 012005. https://doi.org/10.1088/1757-899X/364/1/012005.

Jouan, P., Hallot, P., 2019. Digital twin: a HBIM-based methodology to support preventive conservation of historic assets through heritage significance awareness, Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci. XLII-2/W15, 609–615. https://doi.org/10.5194/isprs-archives-XLII-2-W15-609-2019.

Lucignano, M., 2021. Comunicare l'assenza. L'Incompiuta di Venosa tra Conservazione E Innovazione. Federico II University Press. https://doi.org/10.6093/978-88-6887-095-9.

Marra, A., Gerbino, S., Greco, A., Fabbrocino, G., 2021. Combining integrated informative system and historical digital twin for maintenance and preservation of artistic assets. Sensors 21, 5956. https://doi.org/10.3390/s21175956.

McNeel Rhinoceros Software (Version 7).

Metashape Agisoft PhotoScan Professional Software (Version 1.7.2 – 2021).

MIDAS FEA NX. User's Guide, getting Started & Tutorials. MIDASIT Co., Ltd.

MIDAS Gen. User's Guide, getting Started & Tutorials. MIDASIT Co., Ltd.

MIT - Ministero delle infrastrutture e dei trasporti Circ. C.S.Ll. 21 gennaio 2019 n. 7. Istruzioni per l'applicazione dell'«Aggiornamento delle "Norme tecniche per le costruzioni"» di cui al decreto ministeriale 17 gennaio 2018, 2019 (in italian).it.

Rocca, I., Forti, I., D'Acunto, G., Saetta, A., 2022. Diagnostica, sperimentazione e metodologia di monitoraggio per la conservazione della facciata della chiesa di Santa Maria di Nazareth a Venezia. In: Proceedings of Scienza e Beni Culturali Conference. Bressanone, Italy.

Rocca, I., Berto, L., D'Acunto, G., Saetta, A., 2024. Seismic vulnerability assessment of venetian facades and artistic assets: historic digital twin approach. J. Build. Eng. 95, 109992. https://doi.org/10.1016/j.jobe.2024.109992.

SABAPV-AS. Faldone n. 4: busta A – Cannaregio – scalzi (Santa Maria di Nazareth). Busta I: Cannaregio – Chiesa Degli Scalzi – Varie (1893-1950).

Settis, S., 2016. If Venice Dies. New Vessel Press, New York, NY, USA. ISBN 1939931371.

Themistocleous, K., Ioannides, M., Agapiou, A., Hadjimitsis, D.G., 2015. The methodology of documenting cultural heritage sites using photogrammetry, UAV and 3D printing techniques: the case study of Asinou church in Cyprus. In: Proceedings of the Third International Conference on Remote Sensing and Geoinformation of Environment, 2015, pp. 16–19.

Themistocleous, K., 2020. The use of UAVs for cultural Heritage and archaeology. In: Hadjimitsis, D., et al. (Eds.), Remote Sensing for Archaeology and Cultural Landscapes. Springer Remote Sensing/Photogrammetry. Springer, Cham.. https://doi.org/10.1007/978-3-030-10979-0_14

Xiaozhu, Q., Xingwang, Z., 2018. Application of digital twin technology in the digital construction of material cultural heritage. Information and Documentation Services 39 (2), 103–111.